

①

If $x \geq 12$ AND $y \geq 12 \Rightarrow x+y$

Else $X - Y$

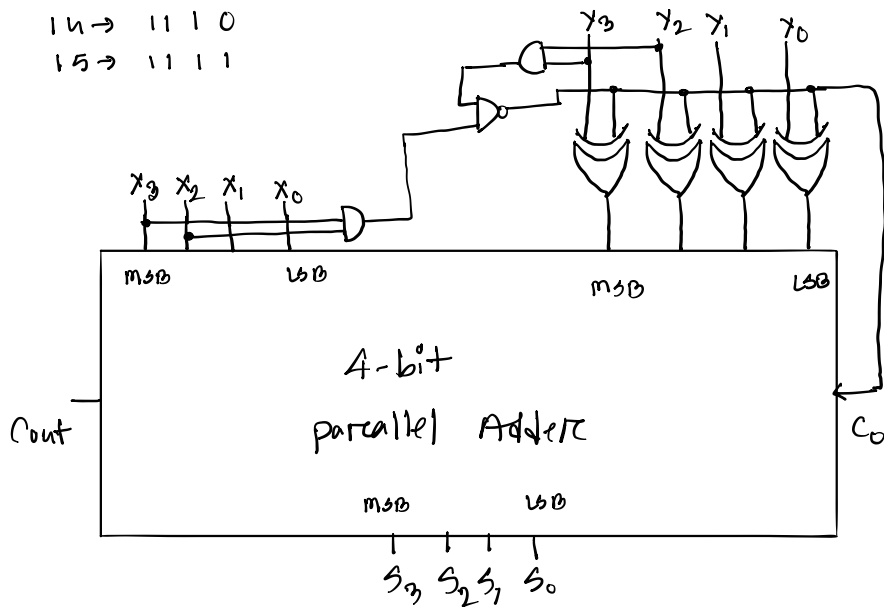
12 → 1100

$13 \rightarrow 1101 \rightarrow \text{Always } 1$

13 → 1101

14 → 11 1 0

15 → 11 11



② Divisible by
 $\hookrightarrow A+B$

↳ $A + B$

Otherwise, $A-B$

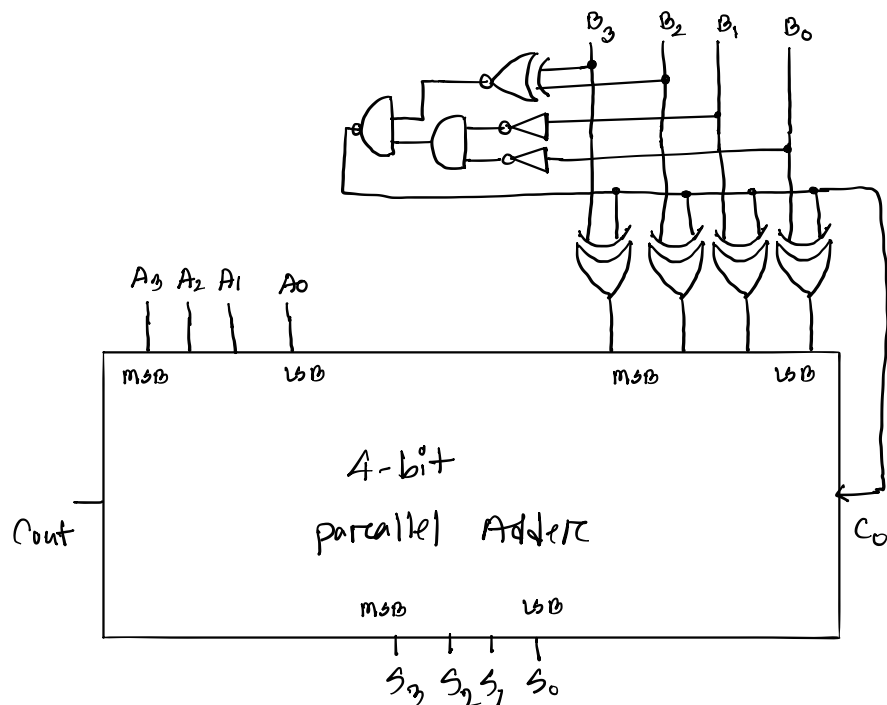
12 $\rightarrow$ Only 0, 12

$$12 \rightarrow 0, 12 \Rightarrow \begin{array}{cccc} & \text{only} & & \\ & & B_3 & B_2 & B_1 & B_0 \\ \hline & & 0 & 0 & 0 & 0 \\ & & 1 & 1 & 0 & 0 \end{array}$$

$$= \overline{b_3} \overline{b_2} \overline{b_1} \overline{b_0} + b_3 b_2 \overline{b_1} \overline{b_0}$$

$$= (B_3 B_2 + \overline{B_3} \overline{B_2}) \quad \overline{B_1} \overline{B_0}$$

$$= (B_3 \oplus B_2) \bar{B}_1 \bar{B}_0$$



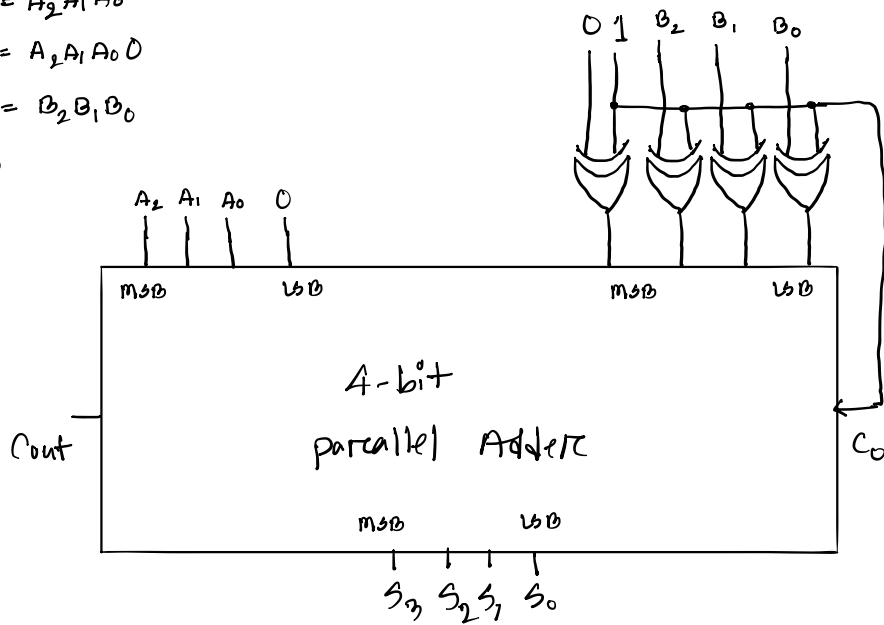
③

$$A = A_2 A_1 A_0$$

$$2A = A_2 A_1 A_0 D$$

$$B = B_2 B_1 B_0$$

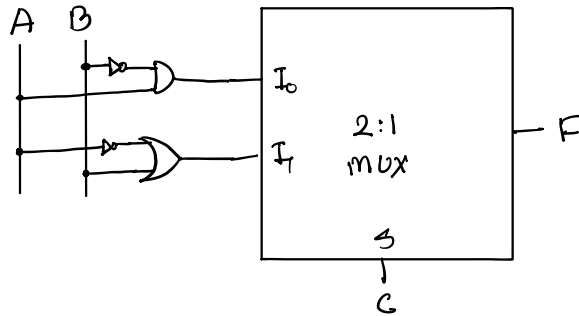
$$\therefore 2A - B$$



④

$$F(A, B, C) = \Sigma(1, 3, 4, 7)$$

	I_0	I_1
$\overline{A}\overline{B}$	0	①
$\overline{A}B$	2	③
$A\overline{B}$	④	5
AB	6	⑦
	$A\overline{B}$	$\overline{A} + B$



⑤

$$F(A, B, C, D) = \Sigma(0, 1, 2, 7, 10, 11, 14)$$

